Mathematical Finance

Labix

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Plan of the notes:

Introduce financial assets

No arbitrage assumption and its mathematical consequences.

Binomial and Black Scholes model.

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1.1 Modeling the Market with Stochastic Processes

A bond is a financial asset in which you lend B_0 to a bank today and received B_T at time $T \in \mathbb{R}_{\geq 0}$. The value of the bond at time $t \in [0, T]$ is given by B_t. The value of the bond is fixed and has no uncertainty. We model this as a stochastic process in the following way.

Definition 1.1.1 (Bonds)
Let $(\Omega, \mathcal{F}, P)$ be a filtered probability space with filtration $\{\mathcal{F}_t \mid t \geq 0\}$. A bond is an adapted stochastic process $B : \Omega \times [0, \infty) \to \mathbb{R}$ be an adapted stochastic process that is a finite variation process.

Recall that an adapted stochastic process is a finite variation process if the following are true.

- $B_0 = 0$ almost surely.
- $t \mapsto B_t$ is a finite variation function almost surely.

Recall that a consequence is that B has no local martingale component: It has no local irreducible randomness.

A stock is a financial asset in which you buy a share in a company at price S_0 today, but the price of the stock in the future $S_t \in \mathbb{R}_{\geq 0}$ are unknown. It depends on earning reports, competitor action, macroeconomic news and other random factors. We define more generally the concept of a risky asset.

Definition 1.1.2 (Risky Assets)
Let $(\Omega, \mathcal{F}, P)$ be a probability space with filtration $\{\mathcal{F}_t \mid t \in [0, \infty)\}$. A risky asset on the space is a semimartingale $S : \Omega \times [0, \infty) \to \mathbb{R}$.

We call them risky because the local martingale component makes the random variable truely random.

Definition 1.1.3 (Market Models)
Let $(\Omega, \mathcal{F}, P)$ be a probability space with filtration $\{\mathcal{F}_t \mid t \in [0, \infty)\}$. A market model on the space consists of a bond $B : \Omega \times [0, \infty) \to \mathbb{R}$ and d risky assets $S_i : \Omega \times [0, \infty) \to \mathbb{R}$ for $1 \leq i \leq d$.

It is important to incorporate bonds when we are modelling stock prices. This is because it is related to the time value of money. Suppose that B_t is the bond at time t if you lend the bank 1 at time $t = 0$. This means the bank has deemed that the value of 1 today is worth the same as B_t at time t in the future. This is precisely the depreciation of money. Indeed when we try to model stock prices, we need to take into account the time depreciation of money, and bonds are precisely the way to represent it.

Definition 1.1.4 (Discounted Values)
Let $(\Omega, \mathcal{F}, P)$ be a probability space with filtration $\{\mathcal{F}_t \mid t \in [0, \infty)\}$. Let $(B, S_1, \dots, S_d)$ be a market model. Define the discounted value of S_i to be

$$\widetilde{S}_i = \frac{S_i}{B}$$

This is the value of the stock when the time value of money is accounted for.

Definition 1.1.5 (Trading Strategies)
Let $(\Omega, \mathcal{F}, P)$ be a probability space with filtration $\{\mathcal{F}_t \mid t \in [0, \infty)\}$. Let $(B, S_1, \dots, S_d)$ be a market model. A trading strategy is a predictable process $\varphi : \Omega \times [0, \infty) \to \mathbb{R}^{d+1}$.

Trading strategies are predictable processes because we would like the trader to only use information strictly from the past.

Definition 1.1.6 (Portfolio Value)
Let $(\Omega, \mathcal{F}, P)$ be a probability space with filtration $\{\mathcal{F}_t \mid t \in [0,\infty)\}$. Let $(B, S_1, \dots, S_d)$ be a market model. Let $\varphi : \Omega \times [0,\infty) \to \mathbb{R}$ be a trading strategy.

- Define the portfolio value of φ to be the stochastic process $V(\varphi) : \Omega \times [0,\infty) \to \mathbb{R}$ given by
$$V = \varphi_0 B + \sum_{i=1}^d \varphi_i S_i$$
- Define the discounted portfolio value of φ to be the stochastic process $\widetilde{V} : \Omega \times [0,\infty) \to \mathbb{R}$ given by
$$\widetilde{V}(\varphi) = \frac{V}{B} = \varphi_0 + \sum_{i=1}^d \varphi_i \widetilde{S}_i$$

Definition 1.1.7 (Admissible Trading Strategies)
Let $(\Omega, \mathcal{F}, P)$ be a probability space with filtration $\{\mathcal{F}_t \mid t \in [0,\infty)\}$. Let $(B, S_1, \dots, S_d)$ be a market model. Let $\varphi : \Omega \times [0,\infty) \to \mathbb{R}^{d+1}$ be a trading strategy. We say that φ is admissible if the following are true.

- φ is $\widetilde{S}$-integrable.
- There exists $a \in [0,\infty)$ such that $\widetilde{V}(\varphi) \geq -a$ almost surely for all $t \in [0,\infty)$.

Definition 1.1.8 (Self-Funding Trading Strategies)
Let $(\Omega, \mathcal{F}, P)$ be a probability space with filtration $\{\mathcal{F}_t \mid t \in [0,\infty)\}$. Let $(B, S_1, \dots, S_d)$ be a market model. Let $\varphi : \Omega \times [0,\infty) \to \mathbb{R}^{d+1}$ be a trading strategy. We say that φ is self financing if
$$\widetilde{V}(\varphi)(-,t) = \widetilde{V}_0(\varphi)(-,t) + \int_0^t (\varphi_1(-,s), \dots, \varphi_d(-,s)) d\widetilde{S}(-,s)$$
almost surely.

An arbitrage opportunity is a trading strategy that

- Requires no money upfront (zero initial cost)
- Never leads to a loss
- Makes a profit with positive probability.

Definition 1.1.9 (Arbitrage)
Let $(\Omega, \mathcal{F}, P)$ be a probability space with filtration $\{\mathcal{F}_t \mid t \in [0,T]\}$. Let $(B, S_1, \dots, S_d)$ be a market model. Let $\varphi : \Omega \times [0,T] \to \mathbb{R}^{d+1}$ be an admissible and self-funding trading strategy. We say that φ is an arbitrage if the following are true.

- $\widetilde{V}(\varphi)(-,0) = 0$ almost surely.
- $\widetilde{V}(\varphi)(-,T) \geq 0$ almost surely.
- $P(\widetilde{V}(\varphi)(-,T) > 0) > 0$.

We assume that arbitrage opportunities do not exist in financial markets.

Definition 1.1.10 (No Free Lunch with Vanishing Risk)
Let $(\Omega, \mathcal{F}, P)$ be a probability space with filtration $\{\mathcal{F}_t \mid t \in [0,T]\}$. Let $(B, S_1, \dots, S_d)$ be a market model. We say that the market satisfies No Free Lunch with Vanishing Risk (NFLVR) if there are no sequence (φ^n) of admissible strategies such that

- $\widetilde{V}(\varphi^n)(-,T) \geq -\frac{1}{n}$.
- $\widetilde{V}(\varphi^n)(-,T)$ converges to X almost surely where X is a random variable such that $P(X > 0) > 0$.

Lemma 1.1.11
Let $(\Omega, \mathcal{F}, P)$ be a probability space with filtration $\{\mathcal{F}_t \mid t \in [0, T]\}$. Let $(B, S_1, \dots, S_d)$ be a market model. If the market satisfies NFLVR, then are no arbitrages in the market.

Theorem 1.1.12 (The Fundamental Theorem of Asset Pricing)

An option is a contract that gives the buyer the right, but not the obligation to buy or sell a stock at a pre-agreed price.

A European call option on a stock S with strike price $K \in \mathbb{R}_{\geq 0}$ and maturity $T \in \mathbb{R}_{\geq 0}$ is a contract that pays its holder

$$\max\{S_T - K, 0\}$$

at time T. It gives the holder the right to buy the stock at price K.

A European put option on a stock S with strike price $K \in \mathbb{R}_{\geq 0}$ and maturity $T \in \mathbb{R}_{\geq 0}$ is a contract that pays its holder

$$\max\{K - S_T, 0\}$$

at time T. It gives the holder the right to sell the stock at price K.

Put-call parity:

$$C_0 - P_0 = S_0 - Ke^{-rT}$$

1.2 The Binomial Model

The one step binomial is a simple model for stock prices and it has two possible futures: the stock either goes up or down in value. The bond grows in a fixed factor.

Definition 1.2.1 (The One Step Binomial)
The one step binomial is the following market model.

- The probability space $\Omega = \{w_u, w_d\}$, $\mathcal{F} = \mathcal{P}(\Omega)$ and $P(\{w_u\}) = p$ and $P(\{w_d\}) = 1 - p$ for some $p \in (0, 1)$.
- The filtration $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_1 = P(\Omega)$.
- The bond $B : \Omega \times \{0, 1\} \to \mathbb{R}$ given by $B_0 = 1$ and $B_1 = 1 + r$ for some $r > 0$.
- The stock $S : \Omega \times \{0, 1\} \to \mathbb{R}$ given by $S_0 > 0$ and $S_1 = uS_0 1_{w_u} + dS_0 1_{w_d}$ where $u > d > 0$.

Notice that S_0 is $\mathcal{F}_1$-measurable but not $\mathcal{F}_0$-measurable: its value at time 1 is not know at time 0, while the bond is $\mathcal{F}_0$-measurable because it is deterministic.

Lemma 1.2.2 (Arbitrage in the One Step Binomial Model)
Let $(\Omega, \mathcal{F}, P, B, S)$ be the one step binomial model given by the above. Let $\varphi = (\varphi^0, \varphi^1)$ be a trading strategy. Then φ is an arbitrage if and only if the following are true.

- $V_0(\varphi) = 0$.
- $V_1(\varphi) \geq 0$.
- $V_1(\varphi)(x) > 0$ for some $x \in \Omega$.

Lemma 1.2.3
Let $(\Omega, \mathcal{F}, P, B, S)$ be the one step binomial model given by the above. Then it has no arbitrage if and only if $d < B_1 < u$.

2 Basic Concepts in Finance

2.1 The Time Value of Money

Concept 2.1.1 (The Time Value of Money)
The time value of money is a core financial principle that states the following. A dollar today is worth more than a dollar in the future.

Most investments are based on this financial principle. There are a few reasons why the time value of money is relevant to the discussion of investment.

- Money can earn interest over time.
- Risk and inflation may reduce future value.
- Investors prefer liquidity.

There are three main types of financial instruments.

Concept 2.1.2 (Financial Instruments)
A financial instrument is a contract that gives rise to a financial asset of one entity and a financial liability or equity instrument of another entity (IAS 32, 39). Some common types of instruments include

- Debt (Fixed Income): A type of investment under which the borrower or issuer is obliged to make payments of a fixed amount on a fixed schedule (e.g. bonds / loans).
- Equity: Equity refers to ownership in an asset or a business (e.g. stocks / shares).
- Derivatives: A derivative is a contract between a buyer and a seller (e.g. options / futures).

Concept 2.1.3 (The Time Value of Money in Fixed Income Instruments)
Let $T \in \mathbb{N}$ be a unit of time (in years). Let FV_t be the future value of an investment at time $t \in [0, T]$. Let PV be the present value of the investment. Assuming the time value of money, let r be the discount rate per period. Then the relationship between the FV_t and PV is described in three different equations depending the type of debt.

- Discount: The investor pays an initial price (PV) and receives a single principal cash flow at maturity (FV). The relation between FV and PV is given by

$$FV_t = PV(1+r)^t$$

- Periodic Interest: The investor pays an initial price (PV) and receives cash flow at fixed time intervals (PMT_{t_i} at time $t_i \in [0, T]$ for $1 \leq i \leq n$), as well as a principal cash flow at maturity (FV). The relation between FV and PV is given by

$$PV = \sum_{i=1}^{n} \frac{PMT_i}{(1+r)^i} + \frac{FV_n}{(1+r)^n}$$

  Perpetual bonds give periodic interest indefinitely with no maturity date. If $PMT_{t_i} = PMT_{t_{i+1}}$ for all i, then we take n to infinity to get

$$PV = \frac{PMT}{r}$$

- Level Payments: The investor pays an initial price (PV) and receives periodic payments at fixed time intervals consisting of principal cash flow and interest. At the end of each period, the annuity cash flow (A_i) is given by the formula

$$A_i = \frac{rPV}{1-(1+r)^{-t_i}}$$

  for time periods $0 < t_1 < \cdots < t_n = T$.

Concept 2.1.4 (The Time Value of Money in Equity)
Let $T \in \mathbb{N}$ be a unit of time (in years). Let FV_t be the future value of an investment at time $t \in [0, T]$. Let PV_t be the present value of the investment at time $t \in [0, T]$. Assuming the time value of money, let r be the discount rate per period. Then the relationship between FV_t and PV is described in three different equations depending the type of equity.

- Constant Dividends: The investor pays an initial price ($PV = PV_0$) and receives a fixed periodic dividend (D). The relation between PV and D is given by

$$PV = \frac{D}{r}$$

- Constant Dividend Growth Rate: The investor pays an initial price ($PV = PV_0$) and receives a periodic dividend (D_{t_i} at time t_i for $1 \leq i \leq n$) with a growth factor of g (so $D_{t_{i+1}} = D_{t_i}(1+g)$). The relation between PV_t and D_t is given by

$$PV_t = \frac{D_{t_i}(1+g)}{r-g} = \frac{D_{t_{i+1}}}{r-g}$$

- Changing Dividend Growth Rate:

If the future value and present value are known, we may solve for r. In this case r is called the implied return instead and it measures how much investors expect to earn from an investment.

Concept 2.1.5 (Cash Flow Additivity Principle)
The cash flow additivity principle is a core financial principle that states the following. The value of a set of cash flow at a given point in time is equal to the sum of values of individual cash flow.

2.2 Return Measures

To assess the performance of an investment, we use a collection of measures that are useful in different situations. They help investors understand how much they have earned relative to the amount invested.

Concept 2.2.1
The rate of return R of an investment is the sum

$$R = \text{Risk Free Interest Rate} + \text{Premium}$$

The risk free interest rate reflects the time preferences of individuals for current versus future real consumption. There are four types of premium depending on the type of cash flow.

- Inflation Premium
- Default Risk Premium
- Liquidity Premium
- Maturity Premium

Concept 2.2.2 (Normalizing Returns)
Return measures are often per annum. Given a rate of return that is not per annum, we can normalize it to obtain a rate of return that is per annum for comparison. If r denotes the quoted rate and is compounded m times per year, then

$$R_{\text{p.a.}} = (1+r)^m - 1$$

If r is quoted per annum but compounded m times, then

$$R_{\text{p.a.}} = \left(1 + \frac{r}{m}\right)^m - 1$$

Concept 2.2.3 (The Different Types of Return Measures)
There are different ways to measure the return of an investment depending on the scenario. Let T be a unit of time (in years). For $t \in [0, T]$, denote P_t the value of the investment at time t. Denote C_t the cash flow towards the investment at end of time t (e.g. $P_0 = -C_0$ indicating that money is spent by the investor to put towards the investment).

- Absolute Return: The naive percentage change that does not account for any other factors
$$R = \frac{P_T - P_0}{P_0}$$
- Holding Period Return: Some financial instruments (such as stocks) provide income just for holding the instrument in the portfolio.
$$R = \frac{P_T - P_0 + I}{P_0}$$
where I denotes the total income. If the investment reports are not annual, we may take the mean (AM, GM, HM) of the rates over a year for ease of comparison.
- The Internal Rate of Return (Money Weighted Rate of Return): Factors in the timing of cash flows towards the investment. The internal rate of return is the percentage return required for the investment to break even with its investment value (usually used for comparing investments) given some prediction of future cash flows. It is given by solving the equation
$$0 = \sum_{t=0}^{n} \frac{C_t}{(1+R)^t}$$
- Time Weighted Rate of Return: Does the opposite of IRR since it exclude the factor of cash flow (usually to evaluate the performance of the investment solely). Suppose that cash flow occurs at discrete times $C_0, \dots, C_{t_n}$ where $t_n = T$. We obtain $n+1$ sub-periods $[0, t_1], \dots, [t_{n-1}, t_n = T]$ in which each sub-period, no cash flow occurs. Let r_i denote the rate of return at the sub-period $[t_{i-1}, t_i]$. If $T = 1$, the time weighted rate of return is given by
$$R = \left(\prod_{i=1}^{n} (1 + r_i) \right) - 1$$
If $T > 1$, denote the time weighted rate of return at the jth year by R_j. Then the overall time weighted rate of return is
$$R = (R_1 \cdots R_T)^{1/T}$$
- Continuously Compounded Return: As m tends to infinity, $\left(1 + \frac{R}{m}\right)^m$ tends to e^R. In this case $P_T = P_0 e^R$. Motivated by this, we define the continuously compounded return to be
$$R = \ln(P_T / P_0)$$

Table for summary:

Return Measure	Time Factor?	Cash Flows?	Best Use Case
Absolute Return	No	No	Simple gains or losses
HPR (Holding Period Return)	Yes	Yes	Specific investment periods
MWRR / IRR (Money-Weighted Return)	Yes	Yes	Actual investor performance
TWRR (Time-Weighted Return)	Yes	No	Comparing fund/manager performance
Log Return (Cont. Compounded)	Yes	No	Quant models, short-term analysis

Table 1: Compact Comparison of Return Measures

Some more specific return measures:
Gross and Net Return
Pre and After Tax Returns

Nominal and Real Returns (Tax related)
Leveraged Returns

3 Debt

4 Equity

5 Derivatives

Assets: Items Stocks: Makes you own a small percentage of a company. Shares: The unit for stocks. (Owning share = Owning a certain number of stocks) Bonds: Buying debt from a loan issuer. Options: Contract that gives the ability to buy / sell assets at a certain time in the future. Futures: Contract to buy / sell assets at a certain time in the future. Value: Total gain / loss from the whole ordeal.

Derivatives: Just think of them as options / future for now. Strike price: the cost of exercising the derivative at the expiration date.

Idea: Derivatives themselves can also be traded.

Call: Looking to buy. Put: Looking to sell.

European options: can only exercise at the specified date. American options: can exercise any time before the specified date.

Derivative Traders: Hedger: Actually exercises the derivative. (To minimize loss) Speculator: Actually bets on the ups and downs of the derivatives. Arbitrageurs: Finding certain entry point to guarantee some winnings by calling / putting a combination of derivatives. (Too powerful and rare so we usually assume no arbitrage opportunities occur)

The price of an option is a function of

- S_0: Current stock price.
- K: Strike price.
- T: Time to expiration.
- r: Risk free interest rate.
- Dividends that are expected to be paid.

More notation:

- S_T: Stock price at expiration day.
- C/P: American call / put.
- c/p: European call / put.

We assume no arbitrage opportunities (if there is then it disappears quickly).

We have

- $C \leq S_0$
- $S_0 - Ke^{rT} \leq c \leq S_0$.
- $P \leq K$
- $Ke^{-rT} - S_0 \leq p \leq Ke^{-rT}$

More:

- $c + Ke^{-rT} = p + S_0$ (idea: European call and put options worth the same price at time T)
- $S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$

6 Mathematical Models for Option Pricing

6.1 Background Terminology

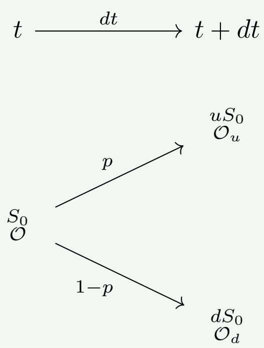
Risk-neutral: A situation where investors do not expect an increase in return when the risk increases. This has two consequences.

- The expected return of a stock is the risk free rate.
- The discount rate for the expected pay off of an option is the risk free.

Risk-less: A portfolio is risk-less when regardless of the outcome, the net total is constant.

6.2 One-Period Binomial Models

Definition 6.2.1 (The Binomial Model) Suppose that t is the current time. An asset has a current stock price of S_0 with an unknown option price $\mathcal{O}$. After a time step of dt, the stock price either increase by a multiplicative factor of $u > 1$ or the stock price decreases by a multiplicative factor of $d < 1$. Denote the corresponding option value (the net gain from the option) by $\mathcal{O}_u$ and $\mathcal{O}_d$ respectively.



The number $0 < p < 1$ is an unknown quantity denoting the probability that the stock price increases.

Lemma 6.2.2 Assume there is no arbitrage opportunity and the world is risk-neutral. Assume the binomial model. Suppose a portfolio of buying $\Delta \in \mathbb{N}$ stocks and shorting a call option. Then the portfolio is risk-less when

$$\Delta = \frac{\mathcal{O}_u - \mathcal{O}_d}{uS_0 - dS_0}$$

Proposition 6.2.3 Assume there is no arbitrage opportunity and the world is risk-neutral. Assume the binomial model. Denote $V(t) = S_0\Delta - \mathcal{O}$ the value of the portfolio of buying Δ stocks and shorting a call option. Then the following are true.

- The probability p that the stock price increases is given by

$$p = \frac{e^{rdt} - d}{u - d}$$

- The option price is given by

$$\mathcal{O} = e^{-rdt}\left(p\mathcal{O}_u + (1-p)\mathcal{O}_d\right)$$

Proof Accounting for the discount rate, we have

$$V(t + dt) = (uS_0\Delta - \mathcal{O}_u)e^{-rdt} = (dS_0\Delta - \mathcal{O}_d)e^{-rdt}$$

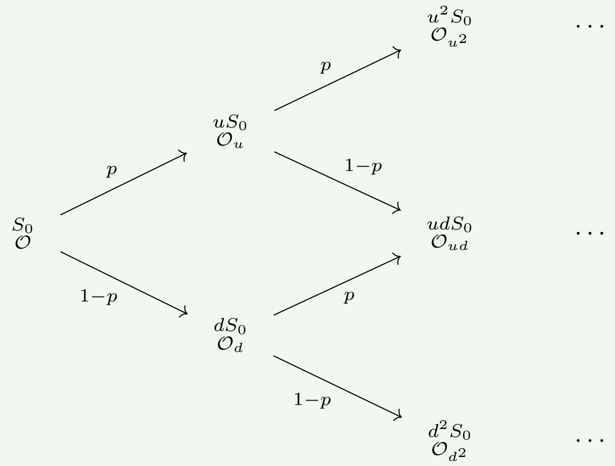
by definition of Δ. Equating $V(t)$ and $V(t + dt)$ gives the desired result. ■

Lemma 6.2.4 Assume there is no arbitrage opportunity and the world is risk-neutral. Assume the binomial model. Suppose that the implied volatility is σ. Then we have

$$u = e^{\sigma\sqrt{dt}} \quad \text{and} \quad d = e^{-\sigma\sqrt{dt}}$$

6.3 Multi-Period Binomial Models

Definition 6.3.1 (The Multi-Period Binomial Model) Suppose that t is the current time. An asset has a current stock price of S_0 with an unknown option price $\mathcal{O}$. After a time step of dt, the stock price either increase by a multiplicative factor of $u > 1$ or the stock price decreases by a multiplicative factor of $d < 1$. Denote the corresponding option value (the net gain from the option) by $\mathcal{O}_u$ and $\mathcal{O}_d$ respectively.



The number $0 < p < 1$ is an unknown quantity denoting the probability that the stock price increases. The process iterates for n-times.

Notice at each time-step, the process is identical except with a different current stock and option price. Therefore we can iterate the one-period binomial model n-times to recover the option value.

Proposition 6.3.2 Assume there is no arbitrage opportunity and the world is risk-neutral. Assume the mutli-period binomial model with period n. Then the European call option price is given by

$$\mathcal{O} = e^{-nrdt}\left(\sum_{i=0}^{n}\binom{n}{i}p^{n-i}(1-p)^i\mathcal{O}_{u^{n-i}d^i}\right)$$

6.4 Black-Scholes-Merton Model

Definition 6.4.1 (Black-Scholes-Merton Model) Let T be a set time (in years). Denote $S : \Omega \times [0, T] \to \mathbb{R}$ the stochastic processes that is the value of a certain stock. Let μ be the annual expected return. Let σ be the annual volatility of the stock price. The Black-Scholes-Merton Model makes the following assumptions

- The percentage in stock is normally distributed:

$$\frac{S_{t+dt} - S_t}{S_t} \sim N(\mu dt, \sigma^2 dt)$$

- μ and σ are constants over time.
- The risk free interest rate is constant over time.
- There are no dividends during the lifetime of the stock.
- There are no arbitrage opportunities.

Lemma 6.4.2 Let T be a set time (in years). Denote $S : \Omega \times [0, T] \to \mathbb{R}$ the stochastic processes that is the value of a certain stock. Let μ be the annual expected return. Let σ be the volatility of the stock price. Assume the Black-Scholes-Merton model. Then we have

$$\ln(S_t) \sim N\left(\ln(S_0) + \left(\mu - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right)$$

Rate of return: The percentage return averaged over a certain amount of time.

Lemma 6.4.3 Let T be a set time (in years). Denote $S : \Omega \times [0, T] \to \mathbb{R}$ the stochastic processes that is the value of a certain stock. Assume the Black-Scholes-Merton model. Let r be the continuously compounded rate of return of S. Then we have

$$r \sim N\left(\mu - \frac{\sigma^2}{2}, \frac{\sigma^2}{T}\right)$$

Volatility: a measure of uncertainty of return. So it is precisely the variance of the rate of return. Hence the annual volatility is σ. Checks out.

Proposition 6.4.4 Let T be a set time (in years). Denote $S : \Omega \times [0, T] \to \mathbb{R}$ the stochastic processes that is the value of a certain stock. Assume the Black-Scholes-Merton model. Then the European call option price according to the model is

$$c = S_0 F_N(d_1) - Ke^{-rT} F_N(d_2)$$

where K is the strike price, r is the risk-free interest rate and F_N is the cumulative distribution function of $N(0, 1)$ and

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \quad \text{and} \quad d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$